Detecting and Attributing AI-Generated Images: A Forensic Fingerprinting Study

A Report Submitted
in Partial Fulfillment of the Requirements
for the Degree of
Bachelor of Technology
in
Computer Science & Engineering

by
Tanish Mittal (20233288)
Vanshika Gautam (20233305)
Shashank Kanaujiya (20233259)
Utkarsh Raj Singh (20233298)
Ashish Meena (20223058)

to the
COMPUTER SCIENCE AND ENGINEERING DEPARTMENT
MOTILAL NEHRU NATIONAL INSTITUTE OF TECHNOLOGY
ALLAHABAD PRAYAGRAJ
October, 2026

UNDERTAKING

I declare that the work presented in this report titled "*Detecting and Attributing AI-Generated Images: A Forensic Fingerprinting Study*", submitted to the Computer Science and Engineering Department, Motilal Nehru National Institute of Technology Allahabad, Prayagraj, for the award of the ***Bachelor of Technology*** degree in ***Computer Science & Engineering***, is my original work. I have not plagiarized or submitted the same work for the award of any other degree. In case this undertaking is found incorrect, I accept that my degree may be unconditionally withdrawn.

October, 2026
Allahabad

(Tanish Mittal (20233288)
Vanshika Gautam
(20233305)
Shashank Kanaujiya
(20233259)
Utkarsh Raj Singh
(20233298)
Ashish Meena (20223058))

CERTIFICATE

Certified that the work contained in the report titled "*Detecting and Attributing AI-Generated Images: A Forensic Fingerprinting Study*", by *Tanish Mittal (20233288)*
Vanshika Gautam (20233305)
Shashank Kanaujiya (20233259)
Utkarsh Raj Singh (20233298)
Ashish Meena (20223058), has been carried out under my supervision and that this work has not been submitted elsewhere for a degree.

(Dr. Anuja Dixit)
Computer Science and Engineering Dept.
M.N.N.I.T, Allahabad

October, 2026

Abstract

AI image generators now produce photorealistic images, which raises a forensic question: was an image captured by a camera, or produced by a generative model, and if so, by which one? This project treats AI-generated image detection as a *source identification* problem. The generator plays the role that a camera plays in classical media forensics, and it is expected to leave its own statistical fingerprint in the pixels. We evaluate detectors in four stages: real-versus-AI detection, generalization to generators unseen during training, robustness to common processing (JPEG compression, resizing, cropping) and attribution to a specific generator.

A pilot study on our own images shows why careful dataset design is essential. Real and AI images could be separated with 100% accuracy using file properties alone, without any pixel analysis. After normalization, every such shortcut fell to chance level (50%).

Acknowledgements

We thank our mentor, Dr. Anuja Dixit, for guidance throughout this project, and the Department of Computer Science and Engineering, MNNIT Allahabad, for its support.

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Chapter 1

Introduction

1.1 Motivation

Online platforms, newsrooms and courts increasingly need to know where an image came from. Generative models such as diffusion models now produce images that are hard to tell apart from photographs by eye.

Metadata cannot answer this question reliably. In our pilot data, every real iPhone photo carried 11 EXIF tags naming the camera, while every AI image carried none. A screenshot or a messaging app strips such metadata from a real photo, and an AI image can be re-saved with any metadata at all. Active approaches, such as watermarks or signed provenance records, work only when the creator cooperates and the record survives sharing. This project therefore follows the *passive forensics* approach: it looks for traces that the image-creation process leaves in the pixel data unintentionally [6].

1.2 Problem Statement

Given only the pixels of an image, decide whether it was produced by a camera or by a generative model and, where possible, identify the generator. The decision should remain reliable for generators not seen during training and after common image processing.

1.3 Research Questions

RQ	Question	Classical forensics analogue
RQ1	Can a detector separate real photographs from AI-generated images from known generators?	Was this image captured by a camera?
RQ2	Does it still work on generators never seen in training?	A camera missing from the reference database
RQ3	How does performance change under JPEG compression, resizing and cropping?	Camera fingerprints weakening after processing
RQ4	Can it identify *which* generator produced an image?	Camera model identification

Table 1: Research questions mapped to classical source identification.

1.4 Contributions

- A bias-controlled evaluation in which real and AI images are matched in file format, resolution and metadata.
- A comparison of a trained CNN detector with a detector built on frozen CLIP features, across known and unseen generators.
- A robustness study under common image processing.
- A test on recent generator output (ChatGPT) that the source dataset does not contain.
- A demo application in which users upload an image and see the detector's prediction.

Chapter 2

Background and Related Work

2.1 Digital Image Forensics

Every step that creates or processes an image leaves statistical traces. A *trace* is any such footprint; a *fingerprint* is a trace that is consistent across all images from one source, distinctive between sources and reasonably stable under normal processing. In a camera, the lens, colour filter array, sensor, demosaicing, in-camera processing and JPEG compression each leave characteristic traces.

The best-known fingerprint is the photo-response non-uniformity (PRNU) of the camera sensor. Lukáš *et al.* [2] extract a noise residual (an image minus its denoised version), average residuals over many images from one camera to obtain a reference pattern and identify the camera of a new image by correlation.

2.2 Generator Fingerprints

A fully AI-generated image never passes through a lens or sensor. Instead, the generator's architecture and training can leave traces of their own. Marra *et al.* [3] showed that GANs leave model-specific fingerprints, in direct analogy with camera PRNU. This analogy is the basis of the project: detecting and attributing AI images is source identification in which the source may be a generative model.

2.3 Learned Detectors

Wang *et al.* [7] trained a standard image classifier (ResNet-50) on images from a single GAN and reported that, with suitable pre-processing and data augmentation, it generalized to other CNN-based generators. Ojha *et al.* [4] observed that detectors trained in this way fail to generalize to newer families of generative models. They proposed classifying images in the feature space of a pre-trained CLIP model that was never trained for real-versus-fake detection.

2.4 Datasets and Dataset Bias

GenImage [8] is a benchmark of about 1.33 million real images (from ImageNet) and 1.35 million images from eight generators, with tasks for cross-generator and degraded-image classification. Grommelt *et al.* [1] showed that GenImage contains JPEG-compression and image-size biases, and that detectors trained on it learn these biases. They reported that removing the biases improved robustness to JPEG compression and raised cross-generator performance by more than 11 percentage points for ResNet-50 and Swin-T detectors.

Chapter 3

Proposed Work

3.1 Overview

The pipeline has five stages: data collection, normalization, splitting, model training and evaluation. Every experiment saves its metrics to a shared results folder, from which the tables in this report and the demo application are produced.

3.2 Dataset

We use Tiny GenImage [5], a scaled-down version of GenImage with 28,000 training and 7,000 validation images. Each image is labelled as real or fake and with its source: Real, ADM, BigGAN, GLIDE, Midjourney, SD14, SD15, VQDM or Wukong.

3.3 Bias Control

- **Format and metadata:** every image is decoded to raw pixels and rebuilt, which removes EXIF, colour profiles and any other embedded data.
- **Resolution:** images are center-cropped to a common size. Cropping is preferred to resizing because resizing alters the fine pixel-level traces that detectors rely on.

- **Compression:** following Grommelt *et al.* [1], both classes are given matched JPEG compression.
- **Splitting:** training, validation and test sets are kept disjoint, and selected generators are held out entirely to answer RQ2.

3.4 Detection Methods

- **M1, trained CNN:** a ResNet-50 pre-trained on ImageNet and fine-tuned for real-versus-fake classification, following the approach of Wang *et al.* [7].
- **M2, frozen CLIP features:** image features from a pre-trained CLIP model with a linear classifier on top, following Ojha *et al.* [4].

3.5 Evaluation Protocol

Results are reported per generator and per processing condition, not only as one overall number, and misclassified images are inspected.

Chapter 4

Experimental Setup and Results Analysis

4.1 Setup

All experiments run on Google Colab with an NVIDIA T4 GPU, using Python, PyTorch, Pillow, NumPy and pandas. Data and results are stored on Google Drive.

4.2 Pilot Study: File-Level Shortcuts (E0)

We collected five real photographs (Apple iPhone 17, HEIC) and five images generated with ChatGPT (PNG). Table 2 shows that the two classes differ on every file-level property.

Property	**Real (5)**	**AI-generated (5)**
Format	HEIF	PNG
Resolution (W×H)	4284×5712	512×512 or 768×512
File size	2,404–5,631 KB	329–524 KB
EXIF tags	11	0
Embedded colour profile	Yes	No

Table 2: File properties of the pilot dataset.

A *shortcut rule* labels an image as AI-generated from one file property, without reading any pixel values. Accuracy is the fraction of images labelled correctly.

4.3 Normalization (E0b)

Each image was rotated upright using its EXIF orientation, converted to RGB, center-cropped to 512×512 pixels, rebuilt from its raw pixel array and saved as PNG. A first attempt without the rebuilding step leaked the iPhone colour profile into all real images, which created a new shortcut. Table 3 compares the rules before and after normalization.

Shortcut rule (predicts "AI" if ...)	**Before (E0)**	**After (E0b)**
File format is PNG	100%	50%
Width under 1,000 pixels	100%	50%
File smaller than 1 MB	100%	50%
No EXIF tags	100%	50%
No embedded information	100%	50%

Table 3: Accuracy of file-level shortcut rules on the pilot set.

Figure placeholder: upload `02_clean_images.png` from the results folder to `figures/`.

Figure 1: Pilot images after normalization. Top: real; bottom: AI-generated.

Two biases remain that file cleaning cannot remove. Real crops show close-up textures, while AI images show centred, well-lit whole scenes (Figure 1). Real pixels also passed through in-camera compression, while AI pixels did not.

4.4 Real versus AI Detection (E1)

To be completed.

4.5 Generalization to Unseen Generators (E2)

To be completed.

4.6 Robustness to Image Processing (E3)

To be completed.

4.7 Generator Attribution (E4)

To be completed.

Chapter 5

Conclusion and Future Work

To be completed.

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